

Shanmathi Gnanabalan

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Master's student in **Environmental & Energy Engineering with a background in Physics**, experienced in **data analysis** roles. Skilled at translating analytical findings into practical sustainability solutions. Passionate about tackling climate change and driving progress towards Net Zero.

Education

MSc Environmental and Energy Engineering | University of Sheffield | Sep 2024 – Present

- Designed air pollution control and CO₂ capture for a 120 MW biomass plant, aligned with UK emission limits and green engineering principles.
- Conducted an **energy audit** of a UK primary school, benchmarking key performance indicators (kWh/m², consumption vs occupancy) to deliver a 32% potential CO₂ reduction.
- **Independently proposed and conducted Environmental Impact Assessment (EIA) and Life Cycle Assessment (LCA)** to evaluate plastic waste in tea bags and recycling strategies.
- Led the grid connection design as a part of a group project on a 40 MW Net-Zero renewable power system in Sheffield, integrating solar, wind, and biomass generation within grid and district heating constraints.
- **Represented MSc cohort** as Academic Representative, communicating student feedback at faculty meetings.
- **Key modules:** Applied Energy Engineering (76%), Low Carbon Energy Science & Technology (75%), Energy Systems & Management (70%), Systems for Sustainability (71%)

MPhys (Hons) Physics with Astrophysics | University of Manchester | Sep 2017 – Jun 2022

- Developed data analysis and problem solving skills through modules in classical and modern physics.
- Achieved 72.5% in the final-year **algorithm design project** by developing a **Python** program to combine and align different medical images. Independently handled data processing, thus improving quantitative and programming ability.
- Achieved **Stellify Award** for leadership and community engagement through volunteering and science outreach.

Work Experience

Data Analysis Intern | University of Manchester | Mar 2023 – Mar 2024

- Produced performance reports for 4 hospital sites, analysing datasets from 2,000+ individuals to identify trends and improvement, primarily using **Excel** and **SPSS**. Used **Power BI** to extract, clean, and analyse data for performance reporting.
- **Presented findings to technical and non-technical stakeholders**, informing policy and operational reviews.
- Designed data collection and storage systems, improving monitoring efficiency and data integrity.
- Awarded **Exceptional Performance Award** for outstanding contribution to improving teaching and learning by providing data driven solutions.

Key Skills

- **Data Analysis & Programming:** Python, gPROMS, Advanced Excel, SPSS, Power BI.
- **Energy & Environmental Engineering:** LCA, EIA, Energy auditing.
- **Research & Modelling:** Algorithm development, optimisation methods, mathematical modelling, technical reporting.
- **Communication & Leadership:** Stakeholder engagement, presenting findings to technical and audiences, student representation, teamwork, volunteering & outreach.
- **Languages:** English (Fluent), Tamil (Native), Hindi (Conversational), French (Beginner).

Projects & Certifications

Net Zero Data Challenge | Yorkshire & Humber Office for Data Analytics | Sep 2025

- Worked as part of a multidisciplinary group analysing emissions and public health data across South Yorkshire to inform regional Net Zero planning.
- **Independently led the health-impact analysis**, analysed emissions & health outcomes across South Yorkshire, benchmarking against Net Zero targets.
- Developed data-informed policy recommendations shared with local councils across South Yorkshire.

Designing Novel Refrigerants for Heat Pumps | MSc Project | May 2025 – Sep 2025

- **Developed and implemented Environmental, Health & Safety (EHS) constraints** within a gPROMS-based optimisation model for industrial heat pumps, integrating flammability, toxicity, and global warming potential into the simulation.
- **Applied mathematical modelling and heat pump thermodynamics** to evaluate heat pump performance and refrigerant behaviour under varying temperature lifts.
- **Independently analysed trade-offs** between energy efficiency (COP) and environmental safety to identify optimal refrigerant candidates for dairy-sector heat pumps.
- Delivered a ranked list of five novel refrigerants, including molecules with no prior documented refrigerant use, demonstrating innovation and scientific rigour.

BCG Climate & Sustainability Job Simulation | Forage Virtual Experience Program | May 2022

- Calculated carbon baselines for a simulated company, and validated against CDP databases.
- Proposed Scope 3 supply chain strategies for Net Zero.

Additional Information

- **Professional Affiliations:** Affiliate Member (Energy Institute), Student Member (ISEP - formerly IEMA), Affiliate Member (IChemE), Member (Institute of Physics).
- **Driving:** Provisional UK licence; lessons in progress (aiming for full licence Feb 2026).